Exploratory Data Analysis Report

Delinquency Prediction Dataset | Summary of Findings & Risk Analysis

This report outlines the primary findings from the exploratory data analysis (EDA) conducted on the delinquency prediction dataset. It highlights underlying data quality issues, tracks missing value imputation strategies, and establishes the foundational risk indicators required for predictive modeling.

1. Key Patterns and Anomalies

- **Low Correlation of Traditional Metrics:** Surprisingly, conventional numeric credit risk markers—such as *Credit Score*, *Credit Utilization*, and historical *Missed Payments*—exhibit near-zero linear correlation (ranging from -0.04 to +0.06) with actual account delinquency.
- **Categorical Inconsistencies:** The *Employment_Status* field contained duplicate representations of identical statuses (e.g., "EMP", "employed", "Employed", "retired"). Standardization was required to prevent algorithms from treating these as distinct variables.
- **Utilization Anomalies:** A small subset of records (4 accounts) showed a *Credit_Utilization* ratio strictly greater than 1.0 (maxing at 1.025). While physically possible (exceeding credit limits), these outliers require winsorization during model training to stabilize linear algorithms.
- **Overall Baseline:** The dataset has an overarching delinquency rate of 16% (80 out of 500 records).

2. Missing Values & Handling Strategy

The dataset contained isolated missing data across three primary financial fields. Industry best practices were applied to resolve these gaps without skewing the underlying statistical distributions.

Feature	Missing Count	Imputation Method Used	Justification
Income	39	Normal Distribution Sampling	Generated synthetic incomes matching the dataset's existing mean ($108k) and standard dev ($53k) to preserve variance, rather than flattening 39 rows with a static average.
Loan_Balance	29	Median Imputation	Loan balances are typically heavily skewed; the median is robust against extreme high-balance outliers.
Credit_Score	2	Median Imputation	With negligible missingness (0.4%), a simple median fill is computationally efficient and introduces zero statistical distortion.

3. Risk Indicators for Delinquency

Given the weak predictive power of the continuous numerical features, the strongest signals for predicting delinquency were found by segmenting the categorical variables. The following high-risk clusters were identified:

Top Categorical Risk Factors

- **Business Credit Cards:** Exhibit the highest product-level risk with a **21.3% delinquency rate**, suggesting significant vulnerability to cash-flow interruptions among this segment.
- **Unemployment Status:** As expected, customers classified as "Unemployed" face the highest default likelihood (**19.4%**), directly linking lack of steady income to missed obligations.
- **Geographic Location (Los Angeles):** Customers residing in Los Angeles show an elevated delinquency rate (**19.6%**) compared to other major cities (e.g., New York at 12.0%).
- **Student Credit Cards:** Represent the second-highest risk tier (**17.9%**), reflecting the financial inexperience and lower incomes typical of this demographic.

4. Strategic Modeling Implications

The EDA dictates a distinct shift in the modeling approach. Because continuous variables provide very little linear signal, standard regression models (e.g., Logistic Regression) are highly likely to underperform.

To accurately predict delinquency with this dataset, machine learning pipelines must heavily weight categorical feature engineering and rely on non-linear, tree-based algorithms—such as **Random Forests** or **Gradient Boosting Machines (XGBoost/LightGBM)**—which can map complex segmentations and interactions.
